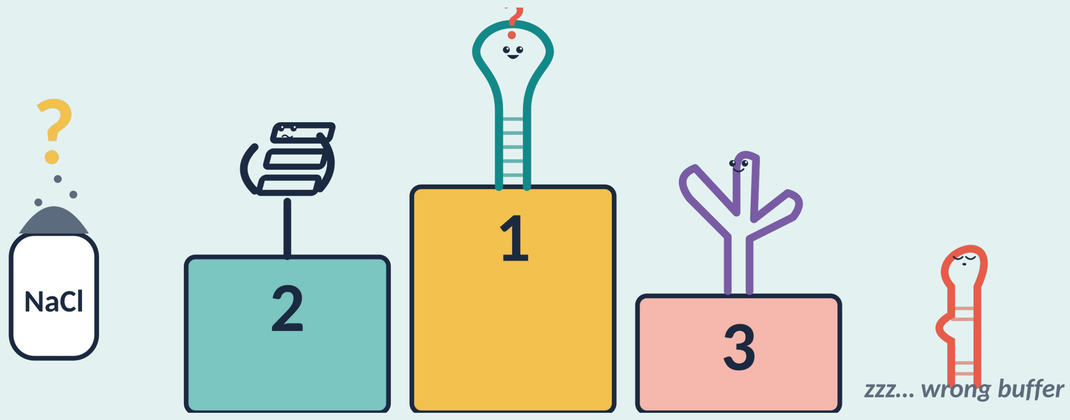
Which aptamer wins?

It depends on the salt.

Why comparing aptamers for rapid virus detection in stool is hard — and how fluorescence polarization helps

[Your name] · [Lab, Institute] · [email] · PhD Retreat 2026

[QR code: talk slides / contact]



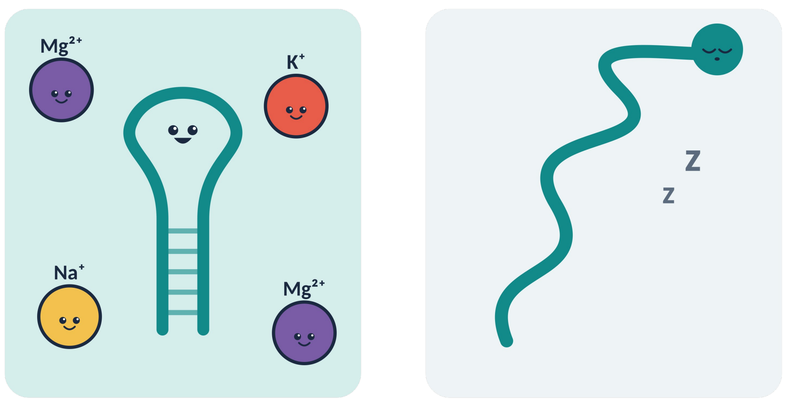
The question

Several published aptamers bind my target virus [name]. Which one should go on the rapid strip test?

The catch

Their ranking changes with salt, method and matrix — so a fair comparison needs a fair method.

1 Aptamers are shape-shifters

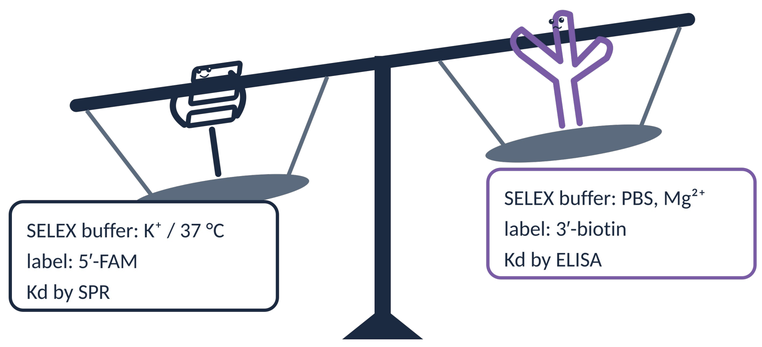


right ions → folded & ready

no Mg^{2+} / wrong pH → floppy

- Mg^{2+} & Na^{+} shield the negative backbone so the strand can fold
- K^{+} specifically stabilises G-quadruplexes; pH & temperature matter too
- No fold → no binding pocket → no binding

2 Apples vs oranges



- Each aptamer was selected (SELEX) in its own buffer
- Labels & linkers differ (FAM, biotin, spacers) and can change binding
- Published Kd values come from different methods → not directly comparable

3 Meet the stool matrix gang



Salt Swing

shifts ionic strength & Mg^{2+} → changes fold



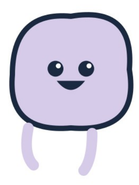
Bile Bubbles

detergent-like bile salts disturb binding



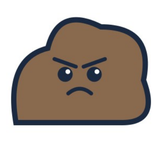
Nuclease Scissors

DNases chew up DNA aptamers



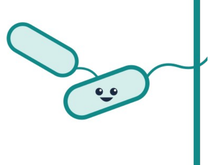
Sticky Proteins

non-specific binding, raise background



Brown Quenchers

humic / pigment colour quench & scatter



Bacteria Crowd

particles, turbidity & competing surfaces

Why it hits every assay

Stool composition varies between donors and days, so salt and Mg^{2+} — and with them the aptamer fold — are never fixed.

Watch-out for optics

Pigments & particles absorb and scatter light — they can fake a signal in colour and fluorescence assays alike.

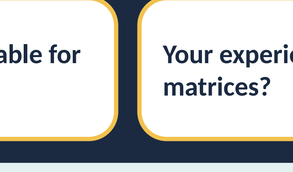
4 ELISA: lessons learned

What we saw	Why it happens
[Weak / patchy signal]	Washes strip aptamers that unbind fast
[Ranking ≠ literature]	Surface tethering changes folding & access
[Result shifts with wash buffer]	One buffer is unfair to aptamers selected in other salts
[High background in stool]	Matrix & blockers stick to plate and DNA

Bottom line: ELISA scored the assay, not the aptamer.

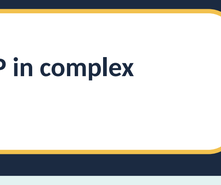
5 Fluorescence polarization to the rescue

Wheee! (fast spin)



light scrambled → LOW polarization

Got it! (slow spin)



light stays aligned → HIGH polarization

- Homogeneous: no surface, no washes
- True equilibrium Kd, minutes per plate
- Same plate, many buffers → fair comparison
- Ratiometric → less sensitive to intensity loss

Planned experiment matrix

	Own SELEX buffer	Shared buffer	NaCl series	Mg^{2+} series	K^{+} (G4)	Stool 1: [x]
Apt-A	●	●	●	●	○	●
Apt-B	●	●	●	●	●	●
Apt-C	●	●	●	●	○	●
Apt-D	●	●	●	●	○	●

Readout → Kd

titrate target, fit mP to a 1:1 binding model

Controls

scrambled sequence · dye only · stool-only blank · unlabelled competitor

Watch-outs

stool scatter/autofluorescence → red dye + blanks · dye/linker can alter binding

Let's discuss!

- **Which buffer is ‘fair’ — each aptamer's own, or one common buffer?**
- **How much stool dilution is acceptable for a rapid test?**
- **Your experience with FP in complex matrices?**

Next (WP3): the winner goes onto a strip

